



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - AAIB-style causal and contributory factors report

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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Synopsis

At 18:47:12 UTC on 11/08/2026 (02:47 local), a 220 t class rigid haul truck descending empty on the main ramp of the Stage 4 open pit at Bindarra Gold Mine struck the right side of a light vehicle emerging from the RL 312 dewatering spur. Within 1.4 seconds the truck operator braked and steered towards the inner windrow, reducing speed from 37 km/h to 24 km/h at impact. The light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the windrow.

Its sole occupant was a grade control technician employed by a survey subcontractor, on his third night shift at the mine. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He remained conscious, was extricated through the rear of the vehicle 27 minutes after the collision and was flown to the metropolitan trauma centre at 07:40 local. His seatbelt was fastened, and the roll-over protection structure deformed without being breached. The light vehicle was destroyed. The haul truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours.

The investigation identified three causal factors:

- The spur joined the ramp 34 m beyond a crest, giving a descending operator 41 m of sight line against the plan's minimum of 120 m and a stopping distance of 96 m. This very likely contributed.
- It was extremely likely that neither operator was aware of the other until about four seconds remained.
- The light vehicle entered the ramp without stopping or establishing the truck's position, which very likely contributed. A full stop would have given the driver only a few metres more sight line.

The contributory factors were:

- The spur was opened under a works order with no management of change raised, so no referral to the plan, the in-cab map or the CAS geofences was made. This very likely contributed.
- The traffic management plan required positive calls only at nine listed intersections, a list reissued once a year. This very likely contributed.
- Neither operator made a call, which likely contributed.
- The S4-LOAD CAS suppression geofence had been extended over the intersection. This likely contributed through the operator's expectation that silence meant an empty road, not through the absent alert itself. The reading that it was not a cause was not excluded.
- The contractor induction was accepted without verification of the site supplement and pit driving assessment. This likely contributed.
- MSG-14 leaves the review trigger undefined and the nomination of interaction zones to the site. This likely contributed, though less firmly than the other factors, and the mine was not in breach of it.

Each contributory factor except the absence of a positive call is a safety issue.

The night-shift emergency response arrangement delayed extrication by approximately 9 minutes. It was unlikely to have affected the occurrence or its outcome, but it very likely increased risk and is a safety issue. The operator's roster placed him on the tenth consecutive night in the circadian low; this was unlikely to have influenced the occurrence, but it likely increased risk. A wet ramp surface and degraded brakes were ruled out as very unlikely. Whether dust reduced the operator's view could not be determined, because nothing at the site measures visibility.

No safety action has been recorded. No safety recommendations are made in respect of the guideline, management of change, radio call or collision awareness issues. [NOT ESTABLISHED: whether any recommendation is to be made in respect of the contractor induction and the emergency response safety issues.]

Factual information

History of the occurrence

Times are UTC unless marked local. The collision occurred at 18:47:12 UTC on 11/08/2026 (02:47 local).

The RL 312 dewatering spur was a 240 m access road cut to dewatering bore DW-14 in the Stage 4 open pit at Bindarra Gold Mine. The works were completed on 31/07/2026. On 02/08/2026 the bore was energised and light vehicles first used the spur. The works log records that the "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed that day and nothing replaced it.

On 04/08/2026 a dewatering technician emerged from the spur in front of a descending haul truck, which he estimated to be about 60 m away. He reversed into the spur and let it pass. He submitted a hazard report at 22:10 local, and the Dewatering Superintendent closed it on 06/08/2026.

On 11/08/2026 dispatch recalled the Stage 4 units after crib at 18:22:41. Haul truck HT-407 completed its first post-crib cycle and dumped its second load. At 18:38:55 the operator reported coming back down.

The grade control technician, driving light vehicle LV-118 for a survey subcontractor, had completed levels at the DW-14 collar and drove out along the spur between 18:44:40 and 18:47:00. He recalled leaving at about ten to three local. He slowed at the spur mouth, looked left up the ramp, saw nothing and drove out at an estimated 25 km/h. On read-back he stated that he did not believe he had come to a complete stop. Nobody on shift knew the light vehicle was in the pit.

HT-407 was descending empty at 37 km/h. The operator saw the beacon whip at about 41 m, applied service brakes and full retard, and steered right towards the windrow. The truck's front left struck LV-118's right side at 24 km/h, 6.2 m into the ramp from the eastern edge of the spur mouth. The light vehicle was pushed 18.4 m down-ramp, overturned onto its left side after 11.2 m and came to rest against the inner windrow facing down-ramp. The truck stopped at 18:47:14.3.

The operator reported the collision at 18:47:41. Dispatch stopped all Stage 4 units at 18:48:01 and told him to remain in the cab, then at 18:48:44 told him the ramp was clear and he could get out. He stayed with the driver, who was conscious. The emergency response team was paged at 02:48 local; the first vehicle arrived at 02:54 local and the second responder, with the hydraulic rescue tool, at 03:07 local. The driver was extricated through the rear at 03:14 local, 27 minutes after the collision, reached the surface medical facility at 03:41 local and was flown to the metropolitan trauma centre at 07:40 local. He had a fractured left clavicle, three fractured left ribs and a small pneumothorax.

The Stage 4 pit was stood down for 9 hours 15 minutes, the ramp below RL 340 closed for 22 hours and the spur barricaded. At 05:10 local the Mining Manager directed a positive pit channel call before entering any Stage 4 intersection, whether or not listed in the traffic management plan.

Recorded information

HT-407's fleet management system (FMS) data were downloaded at 09:30 local by the contractor's technical services group and provided unopened:

Time	Speed	Position and events
18:45:58	41 km/h	RL 368

18:46:31	39 km/h	RL 348, retarder on the second position
18:47:06	37 km/h	RL 314, 41 m above the intersection; last steady data point
18:47:07.4	37 km/h	Service brake pressure rose from 0 to 620 kPa in 0.4 s; retarder to maximum; 14 degrees of right steering
18:47:09.2	32 km/h	—
18:47:12.0	24 km/h	Collision
18:47:14.3	30 km/h	19 m below the intersection

The 1.4 seconds between the last steady point and the first brake pressure rise is taken as the operator's response time.

The collision awareness system (CAS) log recorded a proximity event at 18:47:09.200: tag LV-118 at 21.4 m, closing at 13 km/h, in zone S4-LOAD, with the entry "alert SUPPRESSED (geofence)". No alert was presented in the cab. The event was 2.8 seconds before the collision and 1.8 seconds after braking began. An impact of 1.9 g was logged at 18:47:12.000.

LV-118 carried no FMS; its CAS transponder was a passive tag.

The pit channel recording shows neither vehicle called before the collision. At 18:46:18 dispatch told the excavator operator:

- "407's about four minutes off you."

The operator's first transmission afterwards, at 18:47:41, was:

- "Dispatch, 407. I've hit a light vehicle on the ramp at three-twelve."

Equipment information

HT-407, a 220 t class rigid haul truck, was damaged at the front left bumper, left mudguard and lower two access ladder steps, with nothing above the bumper line. It returned to service on 14/08/2026.

Its CAS operated at Level 7 (operator advisory alerts). Level 9 (automated intervention) is available on this fleet but not enabled at Bindarra. An advisory alert is raised when a tag enters the 60 m detection ring closing above 10 km/h, except inside a suppression geofence.

The FMS in-cab map for Stage 4, last published on 24/07/2026, did not show the spur, so no intersection was displayed at RL 312.

LV-118 was destroyed, with damage concentrated on the right side from the B pillar forward. The roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was fastened and its webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and found 9 m away. The driver stated the beacon was on.

Meteorology

No meteorological observations were obtained. The truck operator described some dust in the air and could not say whether it made a difference. The light vehicle driver described the air as clear, the ramp and windrow plainly visible and no dust in the spur.

Personnel information

Haul truck operator. 11 years in open pit mining, seven on 220 t class trucks, including 14 months at Bindarra with Warrigal Mining Services. Verification of competency issued 06/2025; annual refresher 02/2026; site induction 06/2025. There is no record of a traffic management refresher since. The induction's traffic management module names the nine listed intersections and does not address unlisted ones. He was on night 10 of a 14-night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, having slept 6 hours 30 minutes. Drug and alcohol tests were negative.

Night shift supervisor. 19 years in open pit mining, two years six months in role, working the same pattern as his crews. He was not responsible for the dewatering crew or survey contractor. Drug and alcohol tests were negative.

Light vehicle driver. A grade control technician with Merrivale Survey & Geotechnical with six years' experience, on his third night at Bindarra. He completed the Merrivale two-hour online general mining induction on 24/07/2026 and was issued a Bindarra access card on 09/08/2026 under a recognition agreement after a 20-minute gatehouse briefing. There is no record he completed the site-specific supplement or light vehicle pit driving assessment; the gatehouse required neither. He had driven the spur six or seven times. His radio was on channel 12, as the dewatering foreman instructed. He stated nobody had told him to call before entering the ramp.

No training record exists for anyone on site covering the CAS suppression geofences.

Site information

At RL 312 the ramp runs approximately north-west on a 9.1 per cent descending grade, 24 m between windrows, dry and well maintained. The spur joins from the east on the inside of a left-hand bend at an included angle of 71 degrees, 34 m beyond a crest.

The survey found that a descending operator at 5.80 m eye height first sees the spur mouth at 41 m, and a driver at the spur mouth at 1.45 m eye height first sees a descending truck at 38 m. The 1.6 m inner windrow screens the truck for the last 25 m of the spur approach.

There was no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed for the spur.

The truck's left front tyre scuff began 38.5 m up-ramp of the spur mouth and ran 62 m to its resting position, deviating 1.4 m right over the first 20 m. Glass and mirror fragments from LV-118's right side lay 4.8 m to 14 m below the impact point.

Organisational and management information

Traffic management plan. BIN-TMP-004 provides:

- **Clause 6.2:** intersection sight distance not less than the largest vehicle's stopping distance at the posted speed, and never less than 120 m. An empty 220 t class truck at the posted 40 km/h on a 9 per cent descent needs 96 m. Available sight distance was 41 m.
- **Clause 6.4:** advance warning signs, give-way control and delineation, applying only to intersections on the road layout drawing, which did not show the spur.
- **Appendix B:** lists nine intersections requiring a positive call. The operator and supervisor independently described calling at those nine and nowhere else. The supervisor stated he had never corrected anyone for not calling at an unlisted intersection.
- **Clause 9.1:** out-of-cycle review, never held in the plan's eleven-year history. Annual reviews in 11/2023, 11/2024 and 11/2025 reissued Schedule 1 and Appendix B.

Road changes. Since the 12/11/2025 review, 14 new Stage 4 roads were cut: three under a management of change record, all referred to the plan owner; eleven under a works order, none referred.

Works order WO-26-4471 classified the spur as routine earthworks not requiring management of change under CRL-PRO-022 clause 5.3. It had no scope for signage, survey pickup, plan referral, FMS road network update or geofence review, and its handover field read "N/A". A revised FMS road network is a survey task raised by a management of change record.

CAS suppression geofence. S4-LOAD was drawn on 14/03/2025 around the loading face to stop nuisance alerts. On 09/06/2026 technical services extended it as a configuration task, placing its boundary 180 m up-ramp of the spur, without review against the plan. The supervisor knew alerts were suppressed at the dig but not of the extension. The operator stated he had never received a CAS alert in the pit in 14 months. The standing alert report excludes suppressed events.

Light vehicle entry. There was no permit, call-in, board or pit-entry check for light vehicles.

Emergency response. One mines rescue paramedic is rostered on nights. A second responder, required before any extrication, is called from the accommodation village 11 km from the gatehouse. The post-incident review found extrication was delayed by approximately 9 minutes, without affecting the clinical outcome on this occasion.

Tests and research

HT-407's mean deceleration from brake application to impact was 1.85 m/s^2 , against 1.9 m/s^2 in its monthly brake test on 03/08/2026. Retarder and service brakes performed as specified. Unbraked, the collision speed would have been 37 km/h. LV-118's beacon lamp was tested after recovery and worked.

Other information

Hazard report HAZ-2026-0917 of 04/08/2026 was rated Medium by the technician, who described the rise and windrow, the absent sign and give-way control, and the removed closed sign. He wrote:

- "I do not think they know the road is there."

It was closed on 06/08/2026, rated Low, with a pre-start reminder to the dewatering crew to take care and give way to haul trucks as the only action. It was not referred outside the dewatering department.

A 24-month search found three further reports, from 03/2025, 09/2025 and 02/2026, each describing a light vehicle and haul truck meeting unexpectedly at a different unlisted intersection. Each was closed within three days with a similar reminder, none was referred to the plan owner, and none changed controls.

Analysis

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

The external requirement governing vehicle interaction at Bindarra was Mines Safety Guideline MSG-14 revision 4, made under regulation 6.22 of the Mining Operations Safety Regulations 2018. The regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The guideline is one way of meeting that requirement.

Three of its provisions bear on this occurrence:

- **Review.** Clause 3.2 requires the traffic management plan to be reviewed "at intervals not exceeding twelve months, and following any significant change to the mine's road network". The guideline does not define "significant change" and gives no example of one.
- **Positive communication.** Clause 5.1 requires the operator to nominate the intersections and interaction zones "at which positive communication between vehicles is required before entry". It sets no minimum and states no basis for the nomination.
- **Detectability.** Clause 5.4 requires a light vehicle to be detectable by a heavy vehicle operator. It lists five means, "line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these", and does not say which or how many must be provided.

Clause 3.5 provides that the plan "is a site document" and is not approved by the Inspectorate or submitted to it, except on request after an occurrence or an inspection. It is virtually certain that the guideline left these three terms to the operator. This rests on the documents alone, and their text is not in dispute.

On the day of the occurrence, Bindarra's plan met the guideline on its face:

- it had been reviewed within twelve months;
- it nominated nine interaction zones;
- it provided for detectability by line of sight and by elevated marking.

The mine was not in breach, and nothing in this subsection suggests that the Inspectorate failed in any duty. The matter examined here is narrower. It concerns the shape of the discretion the guideline leaves, and how that discretion was filled at this site.

At Bindarra, all three open terms were filled by one document, BIN-TMP-004, which was reviewed once a year with no outside review.

- **Positive calls.** Clause 4.3.1 requires a positive call only at an intersection "listed at Appendix B". Clause 4.3.2 states that at any other intersection "No call is required."
- **Adding an intersection.** Under the note to Appendix B, an intersection is added only where "it appears on the road layout drawing at Schedule 1" and the annual review assesses it as needing positive communication.
- **Updating the drawing.** Schedule 1 is reissued only at the annual review. No provision allows an intersection to be added to either Schedule 1 or Appendix B between reviews. No provision requires a new road to be referred to the plan's owner when it is cut.

The RL 312 spur was not on the Schedule 1 drawing dated 12/11/2025. As a result, neither the sight distance requirement at clause 6.2 nor the signage, give-way and delineation requirements at clause 6.4 was ever applied to it.

The plan's history shows that this was not confined to one road:

- Annual reviews were completed in each of the three years before the occurrence.
- No review outside the annual cycle had been held under clause 9.1 in the plan's eleven-year life.
- Of the 14 roads cut in the Stage 4 pit since the last review, 11 were cut under a works order and none of those 11 was referred to the plan's owner. The 3 cut under a management of change process were referred.
- A search of the 24 months before the occurrence returned four hazard reports, including HAZ-2026-0917, each describing a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B. Each was closed within three days with a reminder to the crew. None was referred to the plan's owner, and none led to any change to the plan or to a physical control.

The case for influence. The case that the guideline influenced the occurrence rests on this pattern. Each of the three terms the guideline leaves open was filled by the same annual cycle. A defined trigger for review, and a stated minimum or basis for nominating interaction zones, would have given the eleven unreferred roads somewhere to go. On this view, each internal control that did not engage at RL 312 failed for want of exactly such a trigger.

The case against influence. The investigation also weighed a substantial argument against influence. The guideline did not stop anyone from acting:

- Bindarra's own clause 9.1 allowed a review at any time.
- MSG-14 does not prevent a mine from listing every intersection it has.
- The spur was not left out of the plan because the guideline was vague. It was left out because nothing within the mine put the question before the plan's owner.

On that reading, the external provision is background, and the failure lies entirely within the organisation.

How this was resolved. The investigation accepted the influence on balance. It was judged likely that the open terms in MSG-14 influenced the occurrence, by shaping a plan whose only route for recognising a new intersection was the annual review. This is a weaker conclusion than those reached on the factors closer to the collision, and it is not presented as settled.

The argument against retains real force. The organisational factors downstream of the guideline are argued under Organisational Influences and Risk Controls:

- the spur was cut and opened under a works order with no management of change raised;
- positive calls were required only at listed intersections, and the list was rewritten only once a year.

Because the mine could have reviewed its plan or listed the spur at any time without any change to the guideline, removing this factor would not by itself have prevented the collision. The evidence therefore places it as contributory rather than causal.

It is virtually certain that MSG-14 left the review trigger, the nomination of interaction zones and the means of detectability to the operator. It is likely, though less firmly established than the other contributing factors, that this discretion contributed to the occurrence by allowing a new intersection to go unrecognised under a plan that otherwise complied.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

The mine's Management of Change procedure, CRL-PRO-022, is clear about when a record must be raised. Clause 5.2 requires one before "a change to the mine's road network". Clause 5.3 then states that clause 5.2 "does not apply to routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced."

Works order WO-26-4471 cut 240 m of access road to bore DW-14 at RL 312. It classified the work as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3." The Mining Engineer raised the works order and the Mining Superintendent approved it. Its handover field reads "N/A". The procedure, the works order and the survey department's road register are all documents, and they agree with one another. It is virtually certain that the spur was built and opened to traffic without a management of change record.

The procedure itself shows why this mattered. Clause 5.4 places three referrals inside the management of change record:

- to the traffic management plan owner;
- to the survey department for the fleet management system road network;
- to the contractor's technical services group for the collision awareness system configuration.

The works order form at Schedule 4 has no equivalent fields and no referral of any kind. With no record raised, none of the three referrals was made.

The consequences can be traced in each system:

- **The in-cab map.** The fleet management system road network was last published on 24/07/2026, before the spur was cut. On the night of the occurrence it showed the main ramp with no intersection at RL 312.
- **The traffic management plan.** The spur did not appear on the plan's road layout drawing. The signage, give-way and delineation requirements of clause 6.4 therefore did not apply to it.
- **The collision awareness system.** The S4-LOAD geofence had been extended on 09/06/2026. No review of that geofence against the new intersection took place.

The investigation also examined whether this was a one-off. Fourteen new roads had been cut in Stage 4 since the plan's last review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was. The classification on the front page of a works order therefore removed every consequence that should have followed a new road into an operating pit. This had happened eleven times, not once.

The investigation considered whether a referral would necessarily have produced a control, and concluded that it would not. The plan's owner might have considered the spur and still left it unlisted. For this reason the connection to the occurrence is put at very likely rather than higher. That residual doubt also means the absence of a management of change record cannot be treated as a factor without which the collision would not have occurred.

The consequences of the missing referrals are argued elsewhere in the Analysis:

- The sight line at the intersection is argued under Local Conditions.
- The limitation of positive radio calls to listed intersections is argued under Risk Controls.

- The suppression geofence is also argued under Risk Controls.

This section establishes that the clause 5.3 exclusion was the common point through which all three went unaddressed.

It is very likely that opening the spur under a works order, with no management of change raised, contributed to the occurrence by leaving the new intersection absent from the traffic management plan, the in-cab map and the geofence configuration. It was a contributory factor.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

The Contractor Management Standard, CRL-STD-009, sets out how contractor inductions are treated:

- Clause 7.3 allows a contractor's own general induction to be accepted as equivalent where a recognition agreement is held. Merrivale Survey & Geotechnical had held one since 2023.
- Clause 7.4 states that equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted."
- Clause 7.5 makes the engaging department responsible for verifying this before a site access card is issued.

The standard therefore said the right thing. The difficulty lay in how it was applied at the gatehouse:

- The gatehouse issued cards against the recognition agreement alone.
- It held no record of the supplement or the assessment, and was not asked for one.
- Nothing in the access card system distinguished a person who may drive in an operating pit from one who may not.

The verification required by clause 7.5 had no place in which it could happen.

The light vehicle driver's training file shows the following:

- a two-hour Merrivale online module, completed on 24/07/2026;
- a 20-minute gatehouse briefing, covering muster points, speed limits and personal protective equipment;
- a site access card, issued on 09/08/2026;
- no record of the site-specific supplement;
- no record of the pit driving assessment.

The driver had never seen BIN-TMP-004 and did not know it existed. The night of the occurrence was his third night shift at the mine. Merrivale's own records of the site supplement were not available to the investigation. For this reason, it is extremely likely, rather than virtually certain, that the supplement and the assessment were not completed.

The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice. This includes give-way, positive calls and what to do at a blind intersection. Its absence bears directly on how the light vehicle entered the ramp, which is argued under Individual Actions.

Two points weigh against a stronger connection to the occurrence:

- The rule the driver would have been taught, that light vehicles give way at all times, is one he was already applying.
- The sight line at the intersection was capable of defeating an attentive driver on its own.

For these reasons the influence of this factor is put at likely. It cannot be regarded as one whose removal would by itself have prevented the collision. The finding concerns the verification arrangements, not the driver.

It is likely that accepting the contractor's induction as equivalent, without verifying the site supplement and pit driving assessment, contributed to the light vehicle entering the ramp without stopping or establishing the truck's position. It was a contributory factor.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system fitted to HT-407 was serviceable and working as configured. It operated at Level 7, operator advisory, and Level 9 automated intervention, although available on the fleet, was not enabled at Bindarra. An advisory alert is raised when a tag enters the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. The S4-LOAD geofence was drawn on 14/03/2025 around the Stage 4 loading face, where nuisance alerts had made the system unusable. On 09/06/2026 it was extended northward as the face advanced, to a boundary 180 m up-ramp of the RL 312 intersection. The contractor's technical services group made the extension as a system configuration task. It was not recorded in the traffic management plan and was not reviewed against it. The works order for the spur contained no review of the geofences. No training record on site, for any person, covers the geofences.

It is virtually certain that the alert was suppressed. The sealed event log shows a proximity event at 18:47:09.2, with LV-118 at 21.4 m and closing at 13 km/h, marked "alert SUPPRESSED (geofence)". Nothing was presented in the cab.

The evidence against influence is substantial and is set out in full. The suppressed alert would have sounded 1.8 seconds after the operator had begun braking and 2.8 seconds before impact. On this occasion it would have told him something he already knew. The system is advisory only, and with 41 m of sight line the outcome was determined before either the sighting or the alert. On that reading the suppression is a serious matter for the fleet but not a factor in this collision.

The investigation accepted a second reading. It holds that the influence lay in the operator's expectation rather than in the final three seconds. In the twelve months before the occurrence, HT-407 raised 412 advisory alerts, all outside the geofence and none inside it. Over the same period, 68 proximity events inside the geofence were suppressed across the Stage 4 fleet. Nine of those involved a light vehicle tag on the main ramp below the boundary, in the same configuration as this occurrence. Those events were recorded and visible to anyone who queried the system. Nobody did, because the standing report covers alert counts and system health and does not report suppressions. Both the operator and the supervisor raised this matter without being asked. The operator described coming to treat a silent system on that ramp as a clear road, and the statistics are consistent with that account. That expectation was a condition he carried over the crest.

The concern is not that a suppression geofence existed. The concern is that its boundary moved 180 m and nobody outside the group that moved it was informed.

It is likely that the suppression influenced the operator's awareness on the approach. It was a contributory factor rather than a causal one, because an unsuppressed alert would not by itself have prevented the collision.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Clause 4.3.1 of BIN-TMP-004 requires "a positive call ... on the relevant pit channel before entering any intersection listed at Appendix B". Clause 4.3.2 provides that at any other intersection "No call is required." Appendix B lists nine intersections. The note to Appendix B provides that an intersection is added only where it appears on the Schedule 1 road layout drawing and the annual review assesses it as requiring positive communication. Schedule 1 is itself reissued only at that review. No provision allows an intersection to be added between reviews. No provision requires a new road to be referred to the plan's owner. The RL 312 spur appears on neither document. The earliest date on which it could have been listed was 12/11/2026, three months after the collision.

The control was sound in design. Its trigger, however, was circular, and the evidence shows this was not an isolated case. Eleven of the fourteen new Stage 4 roads cut since the last review were never referred to the plan's owner. No out-of-cycle review has been held in the plan's eleven-year history. Four hazard reports in 24 months described light vehicles and haul trucks meeting unexpectedly at unlisted intersections, including HAZ-2026-0917 at this intersection seven days earlier. Each was closed within three days with a reminder to the crew, and none led to a change in the list or to any physical control. The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice: calls were made at the nine intersections on the crib room board and nowhere else.

Had a call been required, the conditions for it to work were in place. The pit channel was open and both vehicles were on it. The dispatcher was tracking the truck closely enough to report "407's about four minutes off you" at 18:46:18. The remaining uncertainty is that a call made but not acknowledged would still have left the driver deciding whether to proceed.

The operators' failure to make a call is analysed under Individual Actions, and the origin of the plan's scope under External Influences. It is very likely that the limitation of positive calls to a list the plan could not update between reviews contributed to the occurrence. On that basis the evidence places this as a causal factor.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The night shift is rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who must be present before any extrication begins, is called in from the accommodation village 11 km from the gatehouse. The emergency response times below are converted from the local times in the response log to the UTC used elsewhere in this report:

- the team was paged at 18:48;
- the first response vehicle reached the scene at 18:54;
- the second responder arrived with the hydraulic rescue tool at 19:07;
- the driver was extricated at 19:14 (03:14 local), 27 minutes after the collision.

The post-incident review found that extrication was delayed by approximately 9 minutes while the team waited for the second responder and the tool.

The collision and the driver's injuries had already occurred before the team was paged. The review recorded that the delay did not affect the clinical outcome, because the driver was conscious and stable throughout. This arrangement therefore played no part in the occurrence or its outcome.

Its importance lies in its structure rather than in this outcome. The driver was restrained and protected by an intact roll-over protection structure. A fuel fire, an unconscious casualty, or an airway needing access within the first minutes would have made the same 9-minute gap decisive.

Nothing in the arrangement scales with the severity of the event it responds to. It is very likely that this single-person night shift extrication capability increases risk to casualties at the site, and it is a safety issue.

Local Conditions

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

The survey of 13/08/2026 placed the spur mouth 34 m beyond the high point of a crest in the main ramp. From the haul truck's eye height of 5.80 m, an operator descending the ramp first had line of sight to the spur mouth at 41 m. From the light vehicle's eye height of 1.45 m, a driver at the spur mouth first had line of sight to a descending truck at 38 m. For the last 25 m of the spur's approach, the 1.6 m inner windrow hid the truck's lower body before that point. The dewatering technician's hazard report of seven days earlier described the same conditions from the same position: the rise in the ramp, the windrow, and emerging onto the ramp before anything could be seen. It is virtually certain that the sight line was as surveyed.

Clause 6.2 of BIN-TMP-004 requires that at every intersection "the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m." For a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade, that stopping distance, calculated with the manufacturer's data and a 1.5 second response, is 96 m. The available 41 m met neither limb of the clause. The intersection also carried no warning sign, no give-way control, no delineation and no posted speed for the spur. Clause 6.4 requires those controls only at intersections shown on the plan's road layout drawing, and the RL 312 spur was not on it.

The consequence follows from the numbers. At the posted speed, the stopping distance was more than twice the distance at which the operator could first see the spur mouth. At 37 km/h, 41 m gave about four seconds before the truck reached the intersection. No level of alertness, training or radio discipline in the cab could have brought the truck to a stand within the distance the geometry allowed. The other matters in this occurrence are all downstream of that geometry:

- the light vehicle entering the ramp without stopping;
- the absence of a positive radio call from either operator;
- the suppressed collision awareness alert.

Each of those, where it was found to contribute, did so by placing a vehicle in a sight line that could not be driven safely. Those matters are argued under Individual Actions and Risk Controls. How the spur came to be opened without a management of change is argued under Organisational Influences.

The sight line on its own injured nobody. It became decisive once another vehicle was in it. With the 120 m that the plan required, the truck's stopping distance would have fallen within the distance available.

It is very likely that the inadequate sight distance at the RL 312 intersection was a causal factor in the collision, because it left a truck at the posted speed unable to stop for a vehicle emerging from the spur.

Dust on the ramp reduced the truck operator's ability to see the light vehicle

The investigation examined whether airborne dust further reduced what the truck operator could see.

Several points supported the possibility:

- The water cart's last pass on the section was at 17:10 UTC, 1 hour 37 minutes before the occurrence. The standing night shift interval is 90 minutes, so a pass was 7 minutes overdue.
- The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 18:40 to 18:50 UTC interval, against a shift mean of 88.
- The truck operator described dust hanging on the ramp, drying out after the cart's earlier pass.
- He was descending behind his own lights into a pit with no fixed lighting, a condition in which airborne dust is at its worst for a driver.

Other points weighed against it:

- The light vehicle driver had been at the same place about four seconds earlier. He was firm that the air was clear, that he could see the ramp surface and the windrow, and that dust at Bindarra was nothing like that at other sites where he worked.
- The truck operator declined to say that the dust had made a difference.
- The monitor reading was well below the site trigger of 250. It measured particulate at a monitor head 180 m away, not what could be seen from either cab. Nothing at Bindarra measures visibility.
- The operator first saw the light vehicle at 41 m, which is the distance the survey shows the geometry gives in clear air. The operator's account and the sight line therefore agree without dust needing to explain anything.

The evidence shows a deviation in the dust conditions that was real and was at the right place and time. It does not establish any effect on detection, in either direction. The two accounts could not be reconciled. Settling the matter would require a measurement of visibility from the cab on the night, and no such measurement exists.

It remains unresolved whether dust was present in a quantity that affected what the operator could see, and no conclusion is drawn from it.

The ramp surface was wet and the truck could not stop in the distance available

A wet or slippery surface was ruled out on three independent sources:

- the weather record showed no precipitation and no rainfall at the site for 26 days;
- the ramp was examined at 06:20 local and found dry throughout;
- the braked tyre scuff ran continuously for 62 m without sign of a locked wheel, and the mean deceleration recorded was 1.85 m/s^2 , against the 1.9 m/s^2 this unit achieved in its dry test.

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

At 02:47 local, the truck operator was 10 hours 47 minutes into the tenth of 14 consecutive night shifts and inside the period of lowest circadian alertness. Every limit in the contractor's fatigue management standard had been met. His own sleep record showed 6 hours 30 minutes of sleep in the preceding day.

The mechanism through which fatigue would have had to act in this occurrence was a slow or absent response. The recorded data contradicts that mechanism:

- the operator responded in 1.4 seconds, within the 1.5 seconds assumed in the plan's own stopping distance calculation;

- the braking and steering that followed were immediate and in the only direction available;
- at 41 m of sight distance, even a faster response would not have produced a different outcome.

It is unlikely that fatigue influenced this occurrence. The exposure nonetheless remains. Every operator on the Stage 4 pit works this roster, and the same roster is worked on three other pits. Each operator drives a 220 tonne vehicle through an unlit pit, where their own eyes are the only defence at an unlisted intersection. That exposure was likely to be an important risk in its own right, and it is recorded as such.

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

The pit channel was logged continuously. The investigation reviewed the whole period from the return from crib at 18:22:41 to two minutes after the collision. No transmission was made by either vehicle about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. Apart from logging on at 17:04, LV-118 made no transmission on the channel at all that shift.

No call was required. BIN-TMP-004 clause 4.3.1 requires a positive call only at the intersections listed at Appendix B. At every other intersection, clause 4.3.2 provides that vehicles give way and states: "No call is required." The RL 312 spur is not listed.

The truck operator, the light vehicle driver and the night shift supervisor were interviewed separately and without prompting. Each described the same practice: calls are made at the nine intersections on the crib room board and at no others. The supervisor stated that he had never corrected anyone for not calling elsewhere, and that requiring calls everywhere would make the channel unusable. The omission therefore belonged to no individual. It matched the plan exactly, and the plan matched the practice. How the plan came to omit the spur is a matter for the analysis of risk controls and organisational influences.

A call from either vehicle would have been heard by the other vehicle, by dispatch and by the supervisor. Dispatch was already following the truck closely enough to tell the excavator operator at 18:46:18 that "407's about four minutes off you." A call from the spur would therefore very probably have been placed against a truck position that dispatch already held. However, a call gives each party knowledge of the other, not separation from it. Even with a call, the two vehicles would still have had to decide who went first across an intersection with 41 m of sight line.

On that basis, the absence of a call was a contributory factor rather than a causal one. It likely contributed to both vehicles arriving at the intersection with neither operator aware of the other.

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

The driver first said he had stopped at the mouth of the spur. He corrected this himself on read-back, against his own interest. His corrected account was that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out. He does not believe he came to a complete stop. The scene evidence is consistent with this account. The impact was 6.2 m into the ramp from the eastern edge of the spur mouth, and the light vehicle's gouge runs from that point. It is virtually certain that the light vehicle entered the ramp in this way.

The driver described applying the rule in clause 4.1, that light vehicles give way to heavy vehicles at all times. At this location, nothing gave that rule any physical form. There was no give-way

control, no stop line, no warning sign and no posted speed at the mouth, because clause 6.4 applies only to intersections shown on the plan's road layout drawing. The driver had not completed the Bindarra pit driving assessment and had never seen the traffic management plan. How that came about is argued under organisational influences.

The survey shows that, from the mouth at the driver's eye height of 1.45 m, a descending truck first came into view at 38 m. The inner windrow hid the truck's lower body for the last 25 m of the approach. A look to the left from a stationary position would therefore have shown him approximately what his look at about 25 km/h showed him, which was nothing. A full stop would have gained a few metres of sight line. It would not have provided the 96 m that the truck needed to stop.

Had the light vehicle not been on the ramp, there would have been no collision. Entering without establishing where the truck was is the step that placed the light vehicle across the path of a descending haul truck. For that reason this action is placed as causal. It very likely contributed to both vehicles arriving at the intersection with neither operator aware of the other. The qualification carries the same weight as the finding: the geometry of the intersection meant that the driver's attention alone, however careful, could not have made the entry safe.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

The fleet management system download records the last steady data point at 18:47:06, at 37 km/h and 41 m above the intersection. From 18:47:07.4, service brake pressure rose from 0 to 620 kPa in 0.4 seconds. In the same movement, the retarder was taken to maximum and 14 degrees of right steering was applied. That is a response of 1.4 seconds, inside the 1.5 seconds allowed in the stopping distance calculation used for the plan's sight distance requirement.

The scene examination corroborated the download. The braked scuff mark deviated 1.4 m to the right over its first 20 m, towards the wall side of the ramp. This was the only direction that did not carry the light vehicle towards the drop.

Impact speed was 24 km/h. With no braking, the download analysis gave 37 km/h. The steering input placed the collision on the wall side of the road, and the light vehicle came to rest against the inner windrow. The collision awareness alert that might otherwise have prompted the operator was suppressed. That matter is argued under risk controls. The operator had begun braking 1.8 seconds before the proximity event was even logged.

This action did not cause the occurrence and did not contribute to it. It very likely reduced the severity of the outcome. It was a significant reason why the injury was survivable.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

Every line of enquiry in this investigation arrives at one condition. At the moment the collision became unavoidable, two vehicles were converging on the RL 312 intersection and neither operator knew the other was there. The convergence is therefore treated here as a single mechanism. It is not five separate findings.

HT-407 was descending the ramp empty at 37 km/h. Its in-cab road network, last published on 24/07/2026, showed no intersection at RL 312. There was no warning sign on the approach and no call on the channel. The collision awareness system logged the light vehicle but presented nothing in the cab.

LV-118 was leaving the spur at about 25 km/h. It had 38 m of sight line to a descending truck and no give-way control at the spur mouth. Its driver made no call and had no knowledge of the truck.

The survey found that the truck operator's first line of sight to the spur mouth was at 41 m. The fleet management system download placed the truck at that point, at 37 km/h, at 18:47:06. It was extremely likely that neither operator was aware of the other before that moment, with about four seconds remaining.

Three independent recordings support this: the pit channel log, the download and the collision awareness system log. The interviews with all three people concerned support it as well. None of these sources contains any indication of earlier awareness. The night shift supervisor did not know a light vehicle was in the pit at all, and there was no permit, call-in or board by which he could have known. The finding is put at extremely likely rather than higher because the light vehicle's movements before the last two seconds rest on the driver's own account alone.

It was extremely likely that the convergence led to the collision. The collision followed from it directly and immediately. At 41 m, nothing between the convergence and the impact was capable of preventing it. The truck operator's braking and steering, discussed under Individual Actions, reduced the impact speed but could not avoid the impact.

The conditions that produced the convergence are argued in their own subsections, and only their bearing on this point is noted here:

- the sight distance at the crest, under Local Conditions;
- the suppression of collision awareness alerts inside the extended geofence, under Risk Controls;
- the absence of a positive call, under Individual Actions;
- the light vehicle's entry onto the ramp, under Individual Actions.

Dust on the ramp was also examined as a possible influence on detection. It could not be resolved, is discussed separately, and nothing from it is carried into this conclusion.

The analysis supports the conclusion that the convergence of the two vehicles, each unseen by the other until about four seconds remained, was a causal factor in the collision.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The download recorded the collision at 18:47:12.0 at 24 km/h, 4.6 seconds after the brake application. The truck's accelerometer logged 1.9 g longitudinal.

The scene examination placed the impact 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the inner windrow toe. Damage to LV-118 was concentrated on the right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The recorded data, the damage pattern and the debris field all agree. It is virtually certain that the impact occurred as described.

The geometry of the intersection determined how the vehicles met. A light vehicle emerging from the spur and a truck descending the ramp meet side-on, so the truck struck the side of the light vehicle and not its front. Once that happened, the outcome depended on three things: the light vehicle's structure, its restraint and the impact speed. The configuration and the speed had largely been settled before either operator had sight of the other. The impact was the mechanism by which the convergence became an injury.

The analysis supports the conclusion that the side impact at 24 km/h was a causal factor in the injuries.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from tyre scuff to body panel gouge, which marks where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. With the door on that side against the ground, extrication was through the rear of the vehicle.

The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The scene measurements, the vehicle examination and the medical evidence agree. It is virtually certain that the roll-over occurred as described and that it was how the collision produced the injuries that made this a serious occurrence.

Two defences worked. The driver's seatbelt was found fastened, with loading marks on the webbing. The roll-over protection structure was deformed inward at the right rear mounting but was not breached. The injuries are those that a side impact and a roll-over produce when the structure holds and the restraint is worn. Had either been absent, the outcome would very likely have been more severe.

The analysis supports the conclusion that the roll-over was a causal factor in the driver's injuries, and that the restraint and the roll-over protection structure limited their severity.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

Degraded braking was ruled out. Brake tests on 03/08/2026 and 14/08/2026 gave mean decelerations of 1.9 m/s² and 1.92 m/s², and the occurrence download gave 1.85 m/s². There was no defect history and the pre-start check recorded no defects, so it was very unlikely that the brakes were degraded.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

Findings

1. It is virtually certain that MSG-14 revision 4 left undefined the significant change to the road network that triggers a review of a site's traffic management plan, and left the nomination of interaction zones requiring positive communication to the site with no minimum and no basis for nomination. It is likely that this contributed to the occurrence by giving the roads cut in Stage 4 between annual reviews no defined route into the plan. This contribution was less firmly established than the others identified, because the guideline did not prevent the mine from holding an out-of-cycle review or from listing every intersection, and an alternative reading places the whole of the failure within the mine. The mine was not in breach of the guideline, and this finding concerns the extent of the discretion it leaves, not any failure by the Inspectorate. (Safety issue)
2. The RL 312 dewatering spur was cut and opened to traffic under works order WO-26-4471, classified as routine earthworks excluded by CRL-PRO-022 clause 5.3, with no management of change raised, and this was virtually certain. It is very likely that this contributed to the occurrence, because none of the three referrals a management of change record would have triggered was made: to the traffic management plan owner, to the survey department for the FMS road network, and to the contractor's technical services group for the CAS geofences. The plan, the in-cab map and the geofence configuration were therefore left unchanged, although a referral would not in itself have ensured that the intersection was listed. (Safety issue)
3. It is very likely that the contractor's general induction was accepted as equivalent under CRL-STD-009 clause 7.3 without verification of the site-specific supplement and the light vehicle pit driving assessment, which clause 7.4 states may not be substituted, and that the access card system had no means of recording either item. This likely contributed by leaving the light vehicle driver without the site's instruction on how pit traffic rules applied, including at a blind intersection. The contribution is qualified in two respects: the give-way rule the assessment would have taught was one he was already applying, and the sight line at the intersection would have defeated attention on its own. (Safety issue)
4. With virtual certainty, the S4-LOAD CAS suppression geofence had been extended on 09/06/2026 to a boundary 180 m up-ramp of the RL 312 intersection, without review against the traffic management plan, so that the proximity event logged at 18:47:09.2 raised no alert in the cab. It is likely that this contributed to the occurrence. The contribution did not come through the absent alert itself, which would have sounded 1.8 seconds after braking had begun. It came through the expectation the truck operator had formed over 14 months without an alert in the pit that silence on the ramp meant an empty road. The alternative reading, that the suppression was a serious deficiency in the fleet's defences but not a cause of this collision, was not excluded. (Safety issue)
5. BIN-TMP-004 required a positive call on the pit channel only at the nine intersections listed at Appendix B, which could be added to only at the annual review through the road layout drawing, and did not require a new road to be referred to the plan's owner. This was virtually certain. It very likely contributed to the occurrence, because the RL 312 intersection was on neither document and the next date on which it could have been listed was 12/11/2026. (Safety issue)
6. It is virtually certain that the night shift emergency response was rostered with one mines rescue paramedic, with the second responder, who must be present before any extrication, called in from

the accommodation village 11 km from the gatehouse. This was unlikely to have influenced the occurrence or its outcome, because the injuries were complete before the team was paged and the approximately 9 minute delay to extrication did not affect the clinical outcome. It is nonetheless very likely that the arrangement increased risk, since the same delay would have a different consequence for a casualty needing early extrication. (Safety issue)

7. The spur joined the ramp 34 m beyond a crest, giving a descending truck operator 41 m of sight line against the plan's minimum of 120 m and against a stopping distance of 96 m for an empty 220 t class truck at the posted 40 km/h. This was virtually certain. It very likely contributed to the occurrence, because at that distance a truck at the posted speed could not stop for a vehicle at the intersection however alert its operator was.

8. The hypothesis that the ramp surface was wet and the truck could not stop in the distance available was examined and ruled out as very unlikely. The ramp was found dry throughout, the braked scuff showed no locked or sliding wheel, and the deceleration achieved was consistent with the unit's tested figure on a dry grade.

9. The truck operator was on night 10 of a 14-night swing, 10 hours 47 minutes into a 12 hour 30 minute shift and in the period of lowest circadian alertness, and this was virtually certain. Fatigue was unlikely to have influenced this occurrence, since his recorded 1.4 second response was inside the 1.5 seconds the plan's own stopping distance calculation allows, and at 41 m of sight line even an instantaneous response would have ended in a collision. It is likely that the roster increased risk as a standing exposure worked by every haul truck operator on the pit.

10. Neither operator made a positive call on the pit channel before entering the intersection, which is virtually certain, and none was required there by BIN-TMP-004 or expected in local practice. This likely contributed to the occurrence. The contribution is qualified because a call would have given each operator knowledge of the other but not separation, and the two vehicles would still have had to resolve priority over 41 m of sight line.

11. With virtual certainty, the light vehicle left the spur and entered the ramp at about 25 km/h without coming to a complete stop and without establishing the truck's position. It is very likely that this contributed to the occurrence. There was no sign, stop line, give-way control or posted speed at the spur mouth, and the survey showed that a full stop would have given the driver only a few metres more sight line, not the 96 m the truck needed.

12. The truck operator braked, took the retarder to maximum and steered right towards the inner windrow within 1.4 seconds of the last steady data point, and this was virtually certain. It very likely reduced the severity of the outcome, lowering the impact speed from 37 km/h to 24 km/h and keeping the light vehicle against the wall side of the ramp.

13. It is very likely that both vehicles arrived at the intersection with neither operator aware of the other until the first sighting available, at about 41 m and about four seconds. It is very likely that this contributed to the occurrence, the collision following directly from that convergence.

14. The haul truck's front left struck the right side of the light vehicle at 24 km/h, which was virtually certain. With virtual certainty, this was the mechanism by which the convergence became an injury.

15. The light vehicle was pushed 18.4 m down the ramp and rolled onto its left side, which was virtually certain. It is virtually certain that this contributed to the driver's injuries. His seatbelt was fastened and loaded, and the roll-over protection structure deformed without being breached.

16. The hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance was examined and ruled out as very unlikely. The deceleration achieved during the occurrence lay between the results of brake tests conducted eight days before and three days after it.

The investigation could not determine whether dust on the ramp reduced the truck operator's ability to see the light vehicle, and no finding is made on it. In favour, a water cart pass was 7 minutes overdue and the fixed dust monitor 180 m up-ramp logged PM10 of 142 micrograms per cubic metre against a shift mean of 88. Against, the light vehicle driver described the air as clear, the truck operator declined to say that dust made a difference, and the reading was well below the site trigger of 250. The sighting at 41 m was also what the geometry gives in clear air. Because nothing at Bindarra measures visibility, the effect of dust on detection could not be established either way.

Causal factors

- The spur joined the ramp beyond a crest, giving a descending truck operator 41 m of sight line against a stopping distance of 96 m. This was virtually certain, and it very likely contributed to the occurrence, because a truck at the posted speed could not stop for a vehicle at the intersection.
- It is virtually certain that the light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, and very likely that this contributed to the occurrence, although a full stop would not have given the sight line the truck needed.
- Both vehicles very likely arrived at the intersection with neither operator aware of the other, and this very likely contributed to the occurrence by leaving about four seconds between the first available sighting and the collision.

Contributory factors

- It is virtually certain that MSG-14 revision 4 left the review trigger undefined and the nomination of interaction zones to the site. It is likely that this contributed to the occurrence, though less firmly than the other factors, and the mine was not in breach. (Safety issue)
- The spur was cut and opened to traffic under a works order with no management of change raised, which was virtually certain, and it is very likely that this contributed by leaving the plan, the in-cab map and the geofence configuration unchanged. (Safety issue)
- It is very likely that the contractor induction was accepted as equivalent without the site supplement and pit driving assessment being verified. This likely contributed to the occurrence, qualified by the fact that the driver was already applying the give-way rule the assessment would have taught. (Safety issue)
- With virtual certainty, CAS alerts were suppressed inside a geofence extended to cover the new intersection. It is likely that this contributed through the expectation it created in the truck operator rather than through the absent alert, and the reading that it was not a cause of this collision was not excluded. (Safety issue)
- It is virtually certain that positive radio calls were required only at the intersections listed at Appendix B, a list reissued once a year, and very likely that this contributed to the occurrence. (Safety issue)
- Neither operator made a positive call on the pit channel before entering the intersection, which was virtually certain. This likely contributed to the occurrence, although a call would have given knowledge rather than separation.

Safety action and recommendations

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14, under which Bindarra's traffic management plan was prepared, leaves three matters to the site:

- It requires the plan to be reviewed after any significant change to the road network, without defining that term.
- It requires the operator to nominate its own interaction zones, without setting a minimum or a basis for the nomination.
- It allows detectability to be met by any one of five means.

The plan is a site document. It is not approved by the Inspectorate, nor submitted to it except on request after an occurrence. It is virtually certain that this discretion existed. Bindarra's plan met the guideline on its face, and the mine was not in breach.

The safety significance lies in the shape of that discretion, not in any failing by the Inspectorate. At Bindarra, all three open terms were filled in by the same document, reviewed once a year, with no outside view of it. Eleven of the fourteen new roads cut in Stage 4 since the last review were never referred to the plan's owner. It was likely that the absence of a defined review trigger and a stated minimum contributed to the occurrence, because nothing required those roads to be brought before the plan. That assessment is less firm than the other contributing factors in this report. The guideline did not prevent an out-of-cycle review or a complete listing of intersections. On that view, the failures within the mine account for the whole of the outcome. The factor is contributory, not causal.

No safety action has been recorded in response to this issue, and no recommendation is made.

Access spur cut and opened to traffic under a works order, with no management of change raised

The access spur to bore DW-14 at RL 312 was cut and opened to traffic under works order WO-26-4471. The works order classified the 240 m of new road as routine earthworks within the approved Stage 4 design. It relied on the exclusion at clause 5.3 of CRL-PRO-022 to record that management of change was not required. It is virtually certain that no management of change record was raised.

Clause 5.4 sets out three referrals: to the traffic management plan owner, to the survey department for the fleet management system road network, and to the contractor's technical services group for the collision awareness system geofences. These referrals arise only from a management of change record, so none of them was made. The in-cab road network published on 24/07/2026 was still current on the night of the occurrence and showed no intersection at RL 312.

The absence of the record very likely contributed to the occurrence. It was the only route by which the new intersection could have reached the traffic management plan, the in-cab map or the geofence configuration. The investigation allowed that a referral is not in itself a control, and that the plan's owner might have considered the spur and still left it unlisted.

The issue extends beyond this intersection. Fourteen new roads have been cut in Stage 4 since the last plan review. The three raised under a management of change record were referred to the plan's owner, and none of the eleven raised under a works order was.

No safety action has been recorded in response to this safety issue.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

The mine's contractor management standard, CRL-STD-009, allows a contractor's own general induction to be accepted as equivalent where a recognition agreement is held. It also expressly excludes the Bindarra site-specific supplement and the light vehicle pit driving assessment from that equivalence, and it makes the engaging department responsible for verifying both before an access card is issued. The standard itself was sound. In practice, however, the gatehouse issued cards against the recognition agreement alone. It held no record of either item and was not asked for one, and nothing in the access card system distinguished a person permitted to drive in an operating pit from one who was not. The verification the standard required therefore had nowhere to take place. This condition extremely likely existed, and it likely contributed to the light vehicle entering the ramp without stopping or establishing the truck's position.

The arrangement allowed a person to drive a light vehicle on an active haul road on his third night at the mine. He had not completed the assessment that the site's own standard says may not be substituted, and no part of the system that admitted him was checking for it. The weakness is not particular to this driver, because cards are issued on the same basis to any contractor holding a recognition agreement.

Safety action taken

No safety action has been recorded in response to this safety issue. The site-wide direction issued after the occurrence required a positive call at every Stage 4 intersection. That direction concerns radio calls and does not address the verification of contractor induction or the pit driving assessment.

Safety recommendations

[NOT ESTABLISHED: whether a recommendation is to be made in respect of this safety issue; none is carried in the source material.]

Positive radio calls required only at listed intersections, and the list is rewritten once a year

The operator's traffic management plan, BIN-TMP-004, required a positive call on the pit channel only before entering one of the nine intersections listed at Appendix B, and required no call anywhere else. An intersection could reach that list only by appearing on the road layout drawing at Schedule 1. Both the drawing and the appendix were reissued only at the annual review. Nothing in the plan allowed an intersection to be added between reviews, and nothing required a new road to be referred to the plan's owner when it was cut. The RL 312 spur appeared on neither document. The earliest date on which it could have been listed was 12/11/2026, three months after the collision.

This was a wider pattern, not an isolated omission. Eleven of the fourteen new Stage 4 roads cut since the last review were never referred to the plan's owner. No review outside the annual cycle had been held in the plan's eleven-year history. The truck operator, the light vehicle driver and the night shift supervisor each independently described the same practice: calls were made at the nine intersections on the crib room board and nowhere else.

The positive call was the mine's own defence against this kind of interaction. At this intersection it was disarmed by the plan's own terms. It was very likely that this arrangement contributed to neither operator making a positive call before entering the intersection.

No safety action has been recorded in response to this safety issue, and no safety recommendation is made.

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on the Stage 4 haul fleet operated at Level 7, operator advisory. It raised alerts when a tag entered the 60 m ring above a 10 km/h closing rate, except inside the S4-LOAD suppression geofence. That geofence was drawn around the loading face on 14/03/2025 to stop nuisance alerts. On 09/06/2026 it was extended northward to a boundary 180 m up-ramp of the RL 312 intersection. The contractor's technical services group made the extension as a system configuration task. It was not recorded in the traffic management plan or reviewed against it, and no person on site had been trained on the geofences. At 18:47:09.2 the system detected the light vehicle at 21.4 m and suppressed the alert.

It is likely that this contributed to the occurrence. The alert itself would have come after braking had begun. The contribution lay in the operator's expectation, built over months of silence on that ramp, that no alert meant an empty road.

The significance of the issue extends beyond this collision. In the twelve months before the occurrence, HT-407 raised 412 advisory alerts, none inside the geofence. Over the same period, 68 proximity events inside it were suppressed, nine involving a light vehicle tag on the main ramp. These events were logged but reported to nobody, because the standing report covered alert counts and system health and not suppressions. The concern is not that a suppression geofence existed. It is that its boundary could move 180 m to take in an active intersection without anyone outside the group that moved it being told.

No safety action has been recorded in response to this safety issue, and no safety recommendation is made.

One emergency response paramedic rostered on night shift, with the second responder called from the village

On night shift at Bindarra, one qualified mines rescue paramedic is rostered, based at the surface medical facility. Any extrication requires a second responder, who is called in from the accommodation village 11 km from the gatehouse. On the night of the occurrence, the first response vehicle reached the scene six minutes after the team was paged. The second responder arrived with the hydraulic rescue tool 13 minutes after that, and the driver was extricated 27 minutes after the collision. The site's post-incident review found that the extrication was delayed by approximately 9 minutes while the team waited for the second responder and the tool.

The collision and the driver's injuries were complete before the emergency response was paged. The investigation found it unlikely that the response time affected this occurrence or its outcome. The post-incident review reached the same view on clinical grounds, because the driver was conscious and stable throughout.

The arrangement nevertheless represents a structural gap in a last line of defence, and the investigation found it very likely to be important to future safety. The gap is present on every night shift, and nothing in it changes with the severity of the casualty. In different circumstances, the

same nine-minute wait could produce a different outcome. Examples include a fuel fire, an unconscious casualty, or an airway requiring the vehicle to be opened within the first ten minutes. No safety action has been recorded in respect of this issue.

Distribution

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Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
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Superintendent Dewatering, Corella Resources	P. Nyaguse